

Yijun YUAN

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Research Interests

- 📖 p -adic Galois representations
- 📖 p -adic transcendental number theory
- 📖 Formalization of arithmetic geometry

Employment

- 2025–2027 📖 **Postdoctoral Fellow, Institute for Theoretical Sciences, Westlake University**
Advisor: Huayi Chen

Professional Position

- 2026– 📖 **Member, Committee on Mathematics and Artificial Intelligence, Chinese Mathematical Society**

Education

- 2022–2025 📖 **Ph.D., Yau Mathematical Sciences Center, Tsinghua University**
Pure Mathematics
Thesis title: *The Field of Norms and $(\varphi, \Gamma_{\mathfrak{F}, K})$ -Modules for p -Adic False Tate Curve Extensions.*
Advisor: Lei Fu / Haoran Wang
- 2018 – 2021 📖 **M.Sc., Shanghai Center for Mathematical Sciences, Fudan University**
Pure Mathematics
Thesis title: *Uniformizer of the False Tate Curve Extension of $\mathbb{Q}_p$.*
Advisor: Shanwen Wang / Quanshui Wu
- 2014 – 2018 📖 **B.S., School of Mathematical Sciences, East China Normal University**
Mathematics and Applied Mathematics

Research Publications

Journal Articles

- 1 **Y. Yuan**, “Formalization of non-archimedean functional analysis 1: Spherically complete spaces,” *Journal of Symbolic Logic*, 2026, To appear.
- 2 S. Wang and **Y. Yuan**, “Hyper-algebraic invariants of p -adic algebraic numbers,” *Canadian Mathematical Bulletin*, 2026, To appear. 🔗 DOI: 10.4153/S0008439526102252
- 3 S. Wang and **Y. Yuan**, “Truncated expansion of ζ_{p^n} in the p -adic mal’cev-neumann field,” *Chinese Annals of Mathematics, Series B*, 2026, To appear.

- 4 S. Wang and Y. Yuan, “Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$ (II),” *International Journal of Number Theory*, vol. 20, no. 01, pp. 1–18, 2024.  DOI: 10.1142/S1793042124500015
- 5 S. Wang and Y. Yuan, “Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$,” *The Ramanujan Journal*, vol. 58, no. 2, pp. 549–595, 2022.  DOI: 10.1007/s11139-021-00473-0

Preprints (all submitted)

- 1 S. Wang and Y. Yuan, *p-adic hahn series with sparse support*, 2026. arXiv: 2607.06944 [math.NT].
- 2 Y. Yuan, *Descent of (φ, τ) -modules in characteristic p* , 2025. arXiv: 2511.17029 [math.NT].
- 3 S. Wang and Y. Yuan, *On the p -adic transcendence of $\sum_{k=1}^{\infty} p^{-1/p^k}$* , 2025. arXiv: 2509.24609 [math.NT].
- 4 Y. Yuan, *Formalization of Harder-Narasimhan theory*, 2025. arXiv: 2509.19632 [math.AG].
- 5 Y. Yuan, *Multivariable period rings of p -adic false tate curve extension*, 2025. arXiv: 2505.24064 [math.NT].

Books & Textbooks

- 1 H. Chen, J. Jiang, S. Wang, L. Xiao, Y. Yuan, and P. Yu, *Formalized Mathematics & Proof Assistant Lean*. 2026, Textbook in preparation.

Talks

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| 2026, Jul. 20th |  <i>p-adic mathematics in Mathlib of Lean 4</i>
<i>Lean 4 Formalization Code Review Training Workshop</i>
<i>Peking University</i> |
| 2026, Apr. 17th |  <i>On the algebraicity of p-adic Hahn series</i>
<i>Jinling Conference on Number Theory and Arithmetic Geometry</i>
<i>Nanjing University</i> |
| 2026, Apr. 16th |  <i>Formalization of Non-Archimedean Functional Analysis</i>
<i>Workshop on Formalized Mathematics</i>
<i>Yau Mathematical Sciences Center, Tsinghua University</i> |
| 2026, Apr. 9th |  <i>Formalization of Non-Archimedean Functional Analysis</i>
<i>School of Mathematical Sciences, East China Normal University</i> |
| 2026, Apr. 7th |  <i>Formalization of Non-Archimedean Functional Analysis</i>
<i>AI for Mathematics @ SIMIS</i>
<i>Shanghai Institute for Mathematics and Interdisciplinary Sciences</i> |
| 2025, Nov. 19th |  <i>Formalization of Harder-Narasimhan Theorey</i>
<i>Shanghai Institute for Mathematics and Interdisciplinary Sciences</i> |
| 2025, Jun. 13th |  <i>Arithmetic of spherically complete valued fields in mixed characteristic</i>
<i>Symposium on Number Theory and Algebraic Geometry of Spring 2025</i>
<i>Capital Normal University</i> |

Awards

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| 2024 |  Outstanding Teaching Assistant in Undergraduate Fundamental Courses, Tsinghua University (Second Class) |
| 2023 |  Scholarship of Yau Mathematical Sciences Center (Second Class) |
| |  Tsinghua University Comprehensive Excellence Scholarship (Second Class) |

Peer Review

2024  Journal of Number Theory

Teaching Experience

2026, Autumn  TA for **Foundation of Algebra and Analysis (I)** at Westlake University.
AI4Math training for undergraduates.

2026, Spring  TA for **Foundation of Algebra and Analysis (II)** at Westlake University.
Formalization (Lean 4) training for undergraduates.

2025, Autumn  TA for **Foundation of Algebra and Analysis (I)** at Westlake University.
Formalization (Lean 4) training for undergraduates.

2024, Autumn  TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

2023, Autumn  TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

2022, Autumn  TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

Skills

Coding  Lean 4, Python (SageMath), $\text{\LaTeX}$, Mathematica, ...